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spotCash!

Placid Express

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Overview

spotCash! is the flagship Money Transfer product offered by Placid Express. Access to the product by correspondents is provided through either an API or by use of the company's dedicated web based platform.

Money Transfer transactions conducted through the spotCash! Product are held at a virtual repository within the company's Global Order Processing Hub (GOPH) platform. Disbursements of transactions are made possible by correspondents utilizing a dedicated Web Services API or by using a Web Based Payment System.

spotCash! differs from other cash based payments in a number of ways. The most important difference lies with how transactions are communicated to correspondents. With spotCash!, the transactions will need to be PULLED instead of Placid Express PUSHING the transactions to the correspondents.

> Prior to further review of the document, please take a look at APPENDIX A for a flowchart illustrating the communication between Placid Express and its correspondent in regards to servicing the spotCash! Product.

Assumptions

The following assumptions are made in regards to the correspondent:

- Correspondent has in place its own (either in-house or vendor developed) web based or other systems deployed throughout its payment points (branches and/or agents)
- Correspondent has the necessary resources/expertise to enable the use of Web Services as a platform for establishing transaction exchange interfaces with its partners
- The API will be implemented at the POS part of its application allowing its tellers to input certain parameters, which will initiate a SOAP call via the API to Placid Express' GOPH.
- Correspondent agrees to use the API in conformance with the API's User Agreement

Scope

This document serves as an implementation reference designed for the correspondent's technical team to enable them to implement the API with the correspondent's remittance application platform.

The API is based on Web Services SOAP 1.1 standard. Most major application frameworks/programming languages such as .NET, JAVA, PHP, C provides facilities for utilizing Web Services.

The major components of the API are:

- Authentication
- Locating Transactions
- Processing Transactions
- Activity Report
- Manage locations

Each of the above contains specific operations that are explained in the proceeding sections.

API Components

As stated in the scope, the 5 major components of the API are Authentication, Locating Transactions, Processing Transactions, and Activity Reports & Managing Locations.

Let's elaborate further on each component:

Authentication

The authentication aspect of the API deals with securely identifying the correspondent. In order to authenticate, a number of parameters must be used with each instance of a request made via the API.

THE Authentication COMPONENT IS USED FOR ALL OPERATIONS of the API and the routine is integrated within each respective operation.

The following parameters are used in the authentication aspect of **every operation**.

XML Elements:

Element Name	Length	Type	Required	Notes
PayeeCode	4	String	YES	Will provide
UserName	10	String	YES	Will provide
AccessCode	SHA1	String	YES	See note 1
TerminalUser	10 (Max)	String	YES	See note 2
ExtraValue	10	String	YES	See note 1

Note 1: Populating Access Code

In order to populate the ***AccessCode*** field, you will need the following:

Password (provided by us)
Extra Value (Provided by us)

Formula for supplying password:

To populate the AccessCode field, use the following formula:

SHA1 (Password)

IMPORTANT: PLEASE be sure to always use lowercase SHA1 values. Take this into consideration when concatenating SHA1 values.

That just about sums up the authentication routine. The above routine would need to be used with each request made via the API.

Note 2: Terminal User

The terminal user is the userid of the teller conducting the payment request. Although not strictly necessary, it is recommended that the actual teller's userid (the same id used to logon to the correspondent's remittance platform) be used instead of a single central userid. Should the correspondent desire, it can also choose to implement/assign separate userids to all users of the API.

The GOPH does NOT authenticate the actual terminal user. It is used as a reference for audit purposes. Should the correspondent wish to implement a per terminal user authentication scheme, a separate lists of userids and associated credentials would need to be exchanged between Placid Express and the correspondent prior to implementation of the API

Locating Transactions

SOAP Operation(s):

<LocateTX>

Request: <FetchTX>
Response: <ShowTX>

Locating transactions are the first part of processing a spotCash! payment. Transactions are located by initializing a call via the API to our GOPH (Global Order Processing Hub). Transactions are basically waiting at the GOPH for you to locate, process and eventually disburse the funds associated with the transaction to the beneficiary.

To locate a transaction, the following elements are used:

Please see next page.

Fetching Transactions <FetchTX>:

Element Name	Length	Type	Required	Notes
PayeeCode	4	string	YES	
UserName	10	string	YES	
AccessCode	SHA1	string	YES	
TerminalUser	10 (Max)	string	YES	
ExtraValue	10	string	YES	
Location	20	string	NO	See note 6
TrackingNum	12	string	YES	See note 3
ReceiverName	20	string	NO	See note 4
ReceiverPhone	12	string	NO	See note 4

Note 3: TrackingNum

Short for Tracking Number. This is the primary identifier used to locate a transaction. The Tracking Number is a 12 digit identifier we use to track all transactions. This identifier is printed on the receipt provided to the remitter as well as sent to the remitter and beneficiary's phone via SMS.

BENEFICIARIES OF A spotCash! PAYMENT WILL ALWAYS PROVIDE THE TRACKING NUMBER TO YOUR TELLERS. THE TRACKING NUMBER IS THE PRIMARY IDENTIFIER USED TO LOCATE/PULL A TRANSACTION.

Note 4: ReceiverName & ReceiverPhone

These 2 elements can be optionally used to cross check the details of a transaction. The decision lies with the correspondent whether to implement its usage, which can add an additional layer of security to transaction verification. Implementing these parameters can allow the correspondent to not populate its system user interfaces unless the details match with the response of LocateTX operation.

Example SOAP Request for <FetchTX>

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ns1="http://api.placid.net/ns1">
  <soapenv:Header/>
  <soapenv:Body>
    <ns1:FetchTX>
      <PayeeCode>TST1</PayeeCode>
      <UserName>TestUser01</UserName>
      <AccessCode>6e2bf9b72183bb518f9133ad4bf5e986cba315ed</AccessCode>
      <TerminalUser>khan01</TerminalUser>
      <ExtraValue> c7988901bc </ExtraValue>
      <Location>?</Location>
      <TrackingNum>595868075797</TrackingNum>
      <ReceiverName>?</ReceiverName>
      <ReceiverPhone>?</ReceiverPhone>
    </ns1:FetchTX>
  </soapenv:Body>
</soapenv:Envelope>
```

Display Transaction <ShowTX>

Once a transaction has been successfully located, the API will present a response with the following elements:

Element Name	Length	Type	Required	Notes
TrackingNum	4	string	N/A	
Token	16	string	N/A	See Note 5
Status	4	string	N/A	See Appendix B
StatusMessage	100	string	N/A	See Appendix B
SenderName	60	string	N/A	
SenderCountry	3	string	N/A	
SenderCity	12	string	N/A	
SenderPhone	12	string	N/A	See note 4
ReceiverName	60	string	N/A	See note 4
ReceiverAddress	60	string	N/A	
ReceiverCountry	3	string	N/A	
ReceiverCity	40	string	N/A	
ReceiverPhone	12	string	N/A	MAY BE NULL
ReceiverMobile	12	string	N/A	MAY BE NULL
ReceiverId	2	string	N/A	MAY BE NULL
ReceiverIdNumber	30	string	N/A	MAY BE NULL
PayoutAmount	17	string	N/A	
PayoutCurrency	3	string	N/A	
ExchangeRate	19	string	N/A	
OrgCurrency	3	string	N/A	
OrgAmount	17	string	N/A	

Note 5: Token

Once a transaction has been successfully located, a response is generated with payload of the transaction detail via the <ShowTX> method. A 7 Digit Token identifier will also be included in the payload. This Token must be used along with other parameters to inform the GOPH that a transaction has been successfully disbursed or rejected. This is accomplished through utilizing the <PayTX> or <RejectTX> operations.

Example SOAP Response <ShowTX>

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Body>
    <ns1:ShowTX xmlns:ns1="http://api.placid.net/ns1">
      <TrackingNum>595868075797</TrackingNum>
      <Token>KPBFTQHC9JM3D8W4</Token>
      <Status>0001</Status>
      <StatusMessage>Transaction Information</StatusMessage>
      <SenderName>TEST TEST</SenderName>
      <SenderAddress>35-01 QUEENS BLVD</SenderAddress>
      <SenderCountry>USA</SenderCountry>
      <SenderCity>LONG ISLAND CITY</SenderCity>
      <SenderPhone>7183923500</SenderPhone>
      <ReceiverName>TEST RECEIVER</ReceiverName>
      <ReceiverAddress>TEST ADDRESS</ReceiverAddress>
      <ReceiverCountry>BGD</ReceiverCountry>
      <ReceiverCity>BARGUNA</ReceiverCity>
      <ReceiverPhone/>
      <ReceiverMobile/>
      <ReceiverId>NONE</ReceiverId>
      <ReceiverIdNumber/>
      <PayoutAmount>1485.00</PayoutAmount>
      <PayoutCurrency>BDT</PayoutCurrency>
      <ExchangeRate>74.8000</ExchangeRate>
      <OrgCurrency>USD</OrgCurrency>
      <OrgAmount>19.85</OrgAmount>
    </ns1:ShowTX>
  </soapenv:Body>
</soapenv:Envelope>
```

Processing Transactions

SOAP Operations:

<PayTX>
Request: <PayTX>
Response: <ShowPayTX>

<RejectTX>
Request: <RejectPay>
Response: <RejectRes>

Once a transaction has been located, the correspondent will have 10 minutes to notify the GOPH of its status. This can be accomplished through the <PayTX> and <RejectTX> operation.

PayTX

After verification of the beneficiary's identification details by comparing it to the details obtained via <LocateTX> operation, the correspondent must invoke the <PayTX> request.

IT IS STRICTLY ADVISED THAT FUNDS BE DISBURSED TO THE BENEFICIARY ONLY AFTER OBTAINING THE REFERENCE NUMBER THROUGH INVOCATION OF THE <PAYTX> REQUEST. THE REFERENCE NUMBER WILL BE PROVIDED THROUGH THE RESPONSE <ShowPayTX>.

<PayTX> XML Elements:

Element Name	Length	Type	Required	Notes
PayeeCode	4	string	N/A	
UserName	10	string	N/A	
AccessCode	SHA1	string	N/A	
TerminalUser	10 (Max)	string	N/A	
ExtraValue	10	string	N/A	
Location	20	string	N/A	See note 6
TrackingNum	12	string	N/A	
Token	16	string	N/A	
ReceiverIdType	12	string	N/A	See Appendix B
ReceiverIdNumber	60	string	N/A	

Note 6: Location

The correspondent must provide us a list of locations (individual agent/branches) that will be processing spotCash! Payments. Once this list is provisioned into our GOPH, the Location ID of the payout branch/agent will need to be populated via the Location element. The Location ID can be the actual ID used on the correspondent's remittance platform.

Example Request : <PayTX>

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ns1="http://api.placid.net/ns1">
  <soapenv:Header/>
  <soapenv:Body>
    <ns1:PayTX>
      <ns1:PayeeCode>?</ns1:PayeeCode>
      <ns1:UserName>?</ns1:UserName>
      <ns1:AccessCode>?</ns1:AccessCode>
      <ns1:TerminalUser>?</ns1:TerminalUser>
      <ns1:ExtraValue>?</ns1:ExtraValue>
      <ns1:Location>?</ns1:Location>
      <ns1:TrackingNum>?</ns1:TrackingNum>
      <ns1:Token>?</ns1:Token>
      <ns1:ReceiverIdType>?</ns1:ReceiverIdType>
      <ns1:ReceiverIdNumber>?</ns1:ReceiverIdNumber>
    </ns1:PayTX>
  </soapenv:Body>
</soapenv:Envelope>
```

<ShowPayTX> XML Elements:

Element Name	Length	Type	Required	Notes
TrackingNum	12	string	N/A	
Token	16	string	N/A	
Status	4	string	N/A	See Appendix D
StatusMessage	100	string	N/A	
RefNumber	10	string	N/A	Can be in correspondent provided format

Example Response : <ShowPayTX>

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Body>
    <ns1:ShowPayTX xmlns:ns1="http://api.placid.net/ns1">
      <ns1:TrackingNum>595868078216</ns1:TrackingNum>
      <ns1:Token>3WYTJX5VPK1U927H</ns1:Token>
      <ns1:Status>0001</ns1:Status>
      <ns1:StatusMessage>Transaction Paid</ns1:StatusMessage>
      <ns1:RefNumber>17</ns1:RefNumber>
    </ns1:ShowPayTX>
  </soapenv:Body>
</soapenv:Envelope>
```

RejectTX

There may arise situations where a transaction cannot be successfully processed for payment due to compliance, failed ID verification, etc. In such cases, the correspondent will need to Reject payout of the transaction by invoking the SOAP operation RejectTX.

Please refer to Appendix B for a list of Reject Reasons and associated codes.

<RejectPay> Elements

Element Name	Length	Type	Required	Notes
PayeeCode	4	string	YES	
UserName	10	string	YES	
AccessCode	SHA1	string	YES	
TerminalUser	10 (Max)	string	YES	
ExtraValue	10	string	YES	
Location	20	string	NO	See note 6
TrackingNum	12	string	YES	
Token	16	string	YES	
RejectReason	3	string	YES	See Appendix C

Example Request :<RejectPay>

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ns1="http://api.placid.net/ns1">
  <soapenv:Header/>
  <soapenv:Body>
    <ns1:RejectPay>
      <ns1:PayeeCode>?</ns1:PayeeCode>
      <ns1:UserName>?</ns1:UserName>
      <ns1:AccessCode>?</ns1:AccessCode>
      <ns1:TerminalUser>?</ns1:TerminalUser>
      <ns1:ExtraValue>?</ns1:ExtraValue>
      <ns1:Location>?</ns1:Location>
      <ns1:TrackingNum>?</ns1:TrackingNum>
      <ns1:Token>?</ns1:Token>
      <ns1:RejectReason>?</ns1:RejectReason>
    </ns1:RejectPay>
  </soapenv:Body>
</soapenv:Envelope>
```

<RejectRes> Elements

Element Name	Length	Type	Required	Notes
TrackingNum	12	string	N/A	
Token	16	string	N/A	
Status	4	string	N/A	See Appendix D for a list of status codes
StatusMessage	100	string	N/A	

Example Response :<RejectRes>

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Body>
    <ns1:RejectRes xmlns:ns1="http://api.placid.net/ns1">
      <ns1:TrackingNum>595868078216</ns1:TrackingNum>
      <ns1:Token>U2D1WHPMEGVZQBS4</ns1:Token>
      <ns1:Status>0001</ns1:Status>
      <ns1:StatusMessage>Reject Accepted</ns1:StatusMessage>
    </ns1:RejectRes>
  </soapenv:Body>
</soapenv:Envelope>
```

Working with Reports

SOAP Operations:

<GetReport>
Request: <ReportReq>
Response: <ReportResp>

We provide a basic reporting functionality that can be utilized to obtain information on transactions that were processed (PAID). Transactions that were cancelled, rejected do not appear in this report.

<ReportReq> Elements

Element Name	Length	Type	Required	Notes
PayeeCode	4	string	YES	
UserName	10	string	YES	
AccessCode	SHA1	string	YES	
TerminalUser	10(Max)	string	YES	Reports will show all transactions. Not just the supplied parameter in this element
ExtraValue	10	string	YES	
DateFrom	8	string	YES	yyyymmdd (string)
DateTo	8		YES	yyyymmdd (string)
RowNumber	3		YES	Number of rows to be returned with the response. **For no limit: populate it with ALL**

Example Request :<ReportReq>

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ns1="http://api.placid.net/ns1">
  <soapenv:Header/>
  <soapenv:Body>
    <ns1:ReportReq>
      <ns1:PayeeCode>?</ns1:PayeeCode>
      <ns1:UserName>?</ns1:UserName>
      <ns1:AccessCode>?</ns1:AccessCode>
      <ns1:TerminalUser>?</ns1:TerminalUser>
      <ns1:ExtraValue>?</ns1:ExtraValue>
      <ns1:DateFrom>?</ns1:DateFrom>
      <ns1:DateTo>?</ns1:DateTo>
      <ns1:RowNumber>?</ns1:RowNumber>
    </ns1:ReportReq>
  </soapenv:Body>
</soapenv:Envelope>
```

<ReportResp> Elements

Element Name	Length	Type	Required	Notes
Status	4	string	N/A	
StatusMessage	100	string	N/A	
RowNumber	3	string	N/A	Total number of rows returned
ReportResult	Array	Array	N/A	

Example Response: <ReportResp>

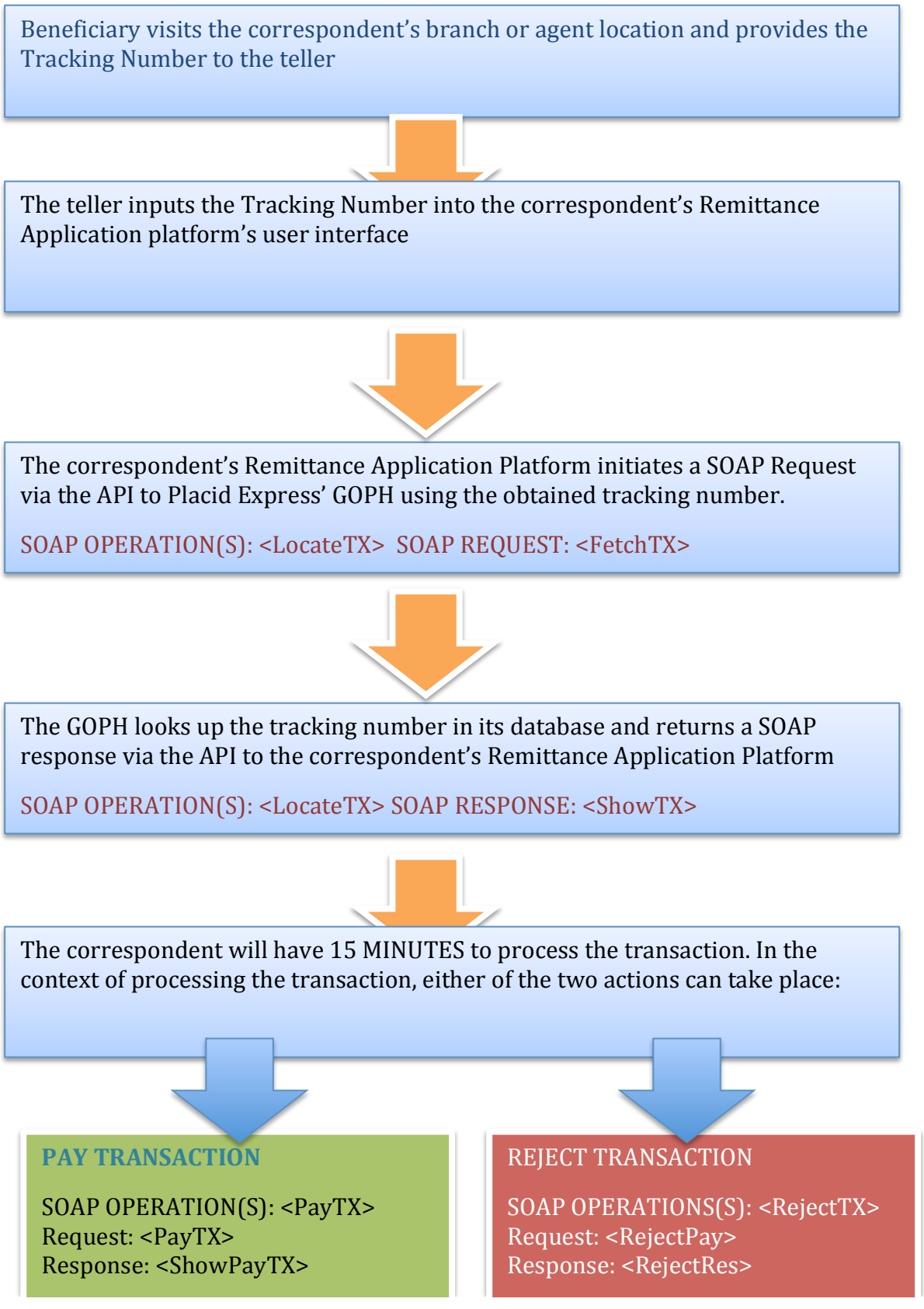
```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Body>
    <ns1:ReportResp xmlns:ns1="http://api.placid.net/ns1">
      <ns1:Status>0001</ns1:Status>
      <ns1:StatusMessage>Report Information</ns1:StatusMessage>
      <ns1:RowNumber>1</ns1:RowNumber>
      <ns1:ReportResult>
        <ns1:ListReport>
          <ns1:PayDate>20110910</ns1:PayDate>
          <ns1:PayTime>091329</ns1:PayTime>
          <ns1:RefNumber>1</ns1:RefNumber>
          <ns1:TrackingNum>591387476715</ns1:TrackingNum>
          <ns1:UserName>dumusrban</ns1:UserName>
          <ns1:TerminalUser/>
          <ns1:SenderName>TEST NEW</ns1:SenderName>
          <ns1:SenderAddress>ANY ADDRESS</ns1:SenderAddress>
          <ns1:SenderCountry>USA</ns1:SenderCountry>
          <ns1:SenderCity>ASTORIA</ns1:SenderCity>
          <ns1:SenderPhone>123456789</ns1:SenderPhone>
          <ns1:ReceiverName>ANY RECEIVER</ns1:ReceiverName>
          <ns1:ReceiverAddress>RECEIVER ADDRESS</ns1:ReceiverAddress>
          <ns1:ReceiverCountry>BGD</ns1:ReceiverCountry>
          <ns1:ReceiverCity>DHAKA</ns1:ReceiverCity>
          <ns1:ReceiverPhone/>
          <ns1:ReceiverMobile/>
          <ns1:ReceiverId/>
          <ns1:ReceiverIdNumber/>
          <ns1:PayoutAmount>0.00</ns1:PayoutAmount>
          <ns1:PayoutCurrency>BDT</ns1:PayoutCurrency>
          <ns1:ExchangeRate>74.8000</ns1:ExchangeRate>
          <ns1:OrgCurrency>USD</ns1:OrgCurrency>
          <ns1:OrgAmount>0.00</ns1:OrgAmount>
        </ns1:ListReport>
      </ns1:ReportResult>
    </ns1:ReportResp>
  </soapenv:Body>
</soapenv:Envelope>
```

Testing Tools

It is recommended that a software-testing package be used in case of issues arising in implementing the API. We have seen good results with the Web Services Testing Framework “SoapUI” which can be obtained from: http://www.soapui.org

Appendix A

Transaction Cycle



Appendix B

List of Identification Documents and codes to be used with the <PayTX> Operation.

IDCode	IDName
LD	DRIVER LICENSE
PA	PASSPORT
RC	NATIONAL ID
VT	VOTER ID

Additional ID codes may be added as per rules of jurisdiction of the correspondent.

Appendix C

Reject Reasons and Codes to be used with the <RejectTX> operation.

RejCode	RejDesc
100	Complience
105	Receiver ID Mismatch
110	Receiver Information Mismatch
115	Fund Shortage
OTH	Other

Appendix D

Various Status messages that are used throughout the API's operations.

StatusCode	StatusMessage
998	Invalid ExtraValue
992	Invalid Payee Code
991	Invalid AccessCode
990	Invalid Tracking Number
993	Invalid TeminalUser
996	Invalid Payee Code
995	Inactive User
994	Unauthorized User
999	Logon Fail
305	Not Available
310	Previously Paid
300	Transaction Not Found
315	Not Ready.
325	Transaction Error
320	Forign Rate Not Found
9910	Invalid UserName
400	Invalid Token
410	Invalid Reject Reason
350	Invalid Date From
351	Invalid Date To
360	Invalid Row Number
0001	**Request Accepted**
0005	No Data Found